Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
3 FT	(a)	(i)		Ammeter in series with correct symbol (1) Voltmeter in parallel with lamp with correct symbol (1)	2			2		2
		(ii)		Use the ammeter to measure current (or amps) (1) Use the voltmeter to measure voltage (or volts) (1) Adjust (or change) variable resistor **and** take values (1)	3			3		3
	(b)			All points correctly plotted ±<1 small square division (2) ignore (0,0) plot 5 points correctly plotted ±<1 small square division (1) ignore (0,0) plot 1-4 points correctly plotted ±<1 small square division (0) ignore (0,0) plot Smooth curve of best fit from (0 – 12 V) ±<1 small square division (1) **Don't accept** thick, double, whispy, wobbly or disjointed curves		3		3	3	3
	(c)	(i)		Reading taken from candidate's graph ± 0.02 shown anywhere - expect 0.62 [A] (1) Correct calculation expect R = 4.84 [Ω] (1) **accept** 4.8 or 5 [Ω] If no workings shown and no line on the graph and answer given between 4.69 and 5 [Ω] award 2 marks		2		2	2	2
		(ii)		[As voltage increases] current increases (1) by decreasing intervals / at a decreasing rate (1) **Alternative:** Gradient decreasing (1) and it is $\frac{1}{\text{resistance}}$ (1) **Alternative:** At least one further calculation of resistance using values from the graph or table (1) with comparison with answer to (c)(i) (1) **Alternative:** When the voltage doubles (1) the increase in current is <u>less than</u> double (1)			2	2		2
				Question 3 total	**5**	**5**	**2**	**12**	**5**	**12**